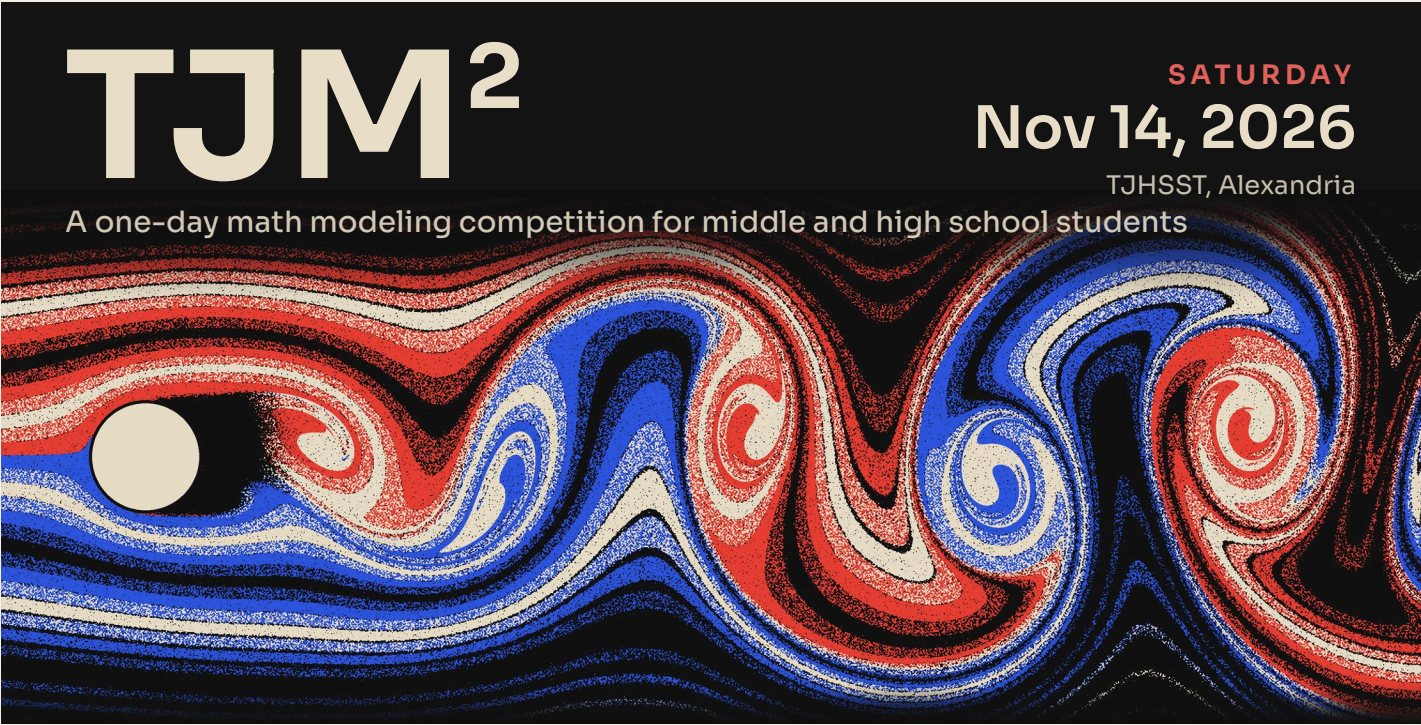




**Fig. 1** Will Northern Virginia’s data centers outgrow the power grid?  
**SAMPLE** The real problem is revealed at 9 a.m. on competition day.

|                                  |                                      |                                      |                              |                              |
|----------------------------------|--------------------------------------|--------------------------------------|------------------------------|------------------------------|
| <b>1 day</b><br>8 a.m. to 7 p.m. | <b>6–12</b><br>grades, two divisions | <b>3–5</b><br>per team, or come solo | <b>\$2,000+</b><br>in prizes | <b>Free</b><br>food included |
|----------------------------------|--------------------------------------|--------------------------------------|------------------------------|------------------------------|

HOW IT WORKS

- 1 Get the problem.**  
Open-ended, built on real data, with no single right answer.
- 2 Build a model with your team.**  
Math, code, spreadsheets: whatever works. Papers are due mid-afternoon.
- 3 Defend it.**  
Finalists present to judges; awards and raffles close the night.

**AI policy.** Use AI to find models and write code. Judges score how well you can justify every decision.

WHY COME

- **\$2,000+ in prizes**, swag for every team that turns in a paper, and raffles.
- **A day out with friends**, with free lunch, dinner and snacks.
- **No experience needed.** We teach the basics that morning and answer questions on our Discord.
- **TJ students:** top performers are considered for our M3 Challenge team.

SCHEDULE

- |              |                          |
|--------------|--------------------------|
| <b>8:00</b>  | Check-in, breakfast      |
| <b>8:45</b>  | Opening ceremony         |
| <b>9:00</b>  | <b>Problem released</b>  |
| <b>12:00</b> | Lunch                    |
| <b>3:00</b>  | <b>Papers due</b>        |
| <b>3:15</b>  | Games while judges read  |
| <b>4:45</b>  | Finalist presentations   |
| <b>6:15</b>  | Dinner                   |
| <b>6:30</b>  | <b>Awards and raffle</b> |

Tentative; details by email.

SAMPLE PROBLEM

Not this year’s

Northern Virginia is the world’s largest data-center market, and Dominion Energy projects its Virginia electricity load could roughly double within 15 years. **(1)** Model a data center’s power draw across a year. **(2)** Forecast the region’s demand 5–15 years out. Will it outgrow supply? Quantify your uncertainty. **(3)** Recommend where the next ones should go.

Data provided: EIA Form 861 · PJM load forecasts · Dominion’s Integrated Resource Plan · JLARC data-center study (2024) · Loudoun County GeoHub. See a winning paper: [TJ’s 2023 M3 Challenge champion](#).



**Sign up to hear more**  
[tjmmc.org/tjm2](https://tjmmc.org/tjm2)

Non-binding. Coming solo? Sign up anyway and we’ll match you with a team.  
Questions: [officers@tjmmc.org](mailto:officers@tjmmc.org)



Hosted by the **TJ Math Modeling Club**  
M3 Challenge champions, 2023 · runners-up, 2024